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Innovative Cooling Solutions for High Intensity Solar Applications

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Abstract

This study presents a comparative experimental analysis of passive and active cooling strategies for thermal management in HCPV modules, focusing on the integration of silver-based nanofluids. The analysis focuses on the use of finned heat sinks and microchannel heat exchangers, employing silver nanofluids as working fluids to improve the thermal and electrical efficiency of these systems under high solar intensity conditions. Two cooling configurations were evaluated: a finned aluminum heat sink (passive) and a microchannel heat exchanger (active), tested under both laboratory and real outdoor conditions. The active system was assessed using distilled water, a water-ethylene glycol mixture, and silver nanofluids, with and without polyvinylpyrrolidone (PVP) stabilization. Results show that while passive cooling offers mechanical simplicity, it fails to effectively dissipate heat under high irradiance, with thermal resistance exceeding 3 °C/W and peak cell temperatures above 66 °C. In contrast, the active microchannel system significantly improved thermal regulation, reducing cell temperatures by more than 17 °C and increasing electrical efficiency up to 29.64%. The use of silver nanofluids further enhanced heat transfer performance, lowering thermal resistance by up to 18.3% and achieving favorable thermohydraulic characteristics. These findings demonstrate that nanofluid-enhanced active cooling systems represent a promising pathway to improve the thermal and electrical performance of HCPV modules, offering scalable and efficient solutions for next-generation solar energy technologies.

Introduction

The increasing urgency to reduce greenhouse gas emissions and transition toward low-carbon energy systems has elevated solar power technologies to the forefront of sustainable development strategies worldwide. Among renewable energy sources, solar energy is particularly appealing due to its abundance, modularity, and potential for distributed generation. Photovoltaic (PV) systems have therefore gained considerable traction; however, their performance remains strongly limited by thermal effects. As a significant portion of the incident solar radiation is not converted into electricity but instead dissipated as heat, elevated operating temperatures lead to decreased electrical efficiency, accelerated material degradation, and shortened module lifespan (Şimşek and Ökten, 2025).

The development of High-Concentration Photovoltaic (HCPV) systems represents a significant leap toward maximizing solar-to-electric conversion efficiency. By concentrating sunlight—often several hundred times—onto compact, high-efficiency multijunction solar cells (e.g., GaInP/GaAs/Ge), HCPV systems can achieve efficiencies exceeding 40%, and up to 47% under controlled laboratory conditions ((NREL), 2019). This improvement, however, comes with a trade-off: the high concentration ratio substantially amplifies the thermal load on the cells. Cell temperatures in excess of 100 °C are frequently observed, which can reduce electrical output, increase thermal stress, and compromise structural integrity over time (Bahaidarah, Baloch, and Gandhidasan, 2016).

To address these challenges, a variety of thermal management techniques have been proposed. Passive cooling solutions—such as extended surface finned heat sinks, phase change materials (PCMs), and natural convection channels—are valued for their simplicity and independence from external energy sources (Royne, Dey, and Mills, 2005). Yet, their thermal dissipation capacity becomes insufficient under the high irradiance levels characteristic of HCPV systems. Active cooling approaches, such as forced air or liquid circulation, offer more effective heat extraction but introduce additional complexity, parasitic power consumption, and potential maintenance concerns (Gilmore, Timchenko, and Menictas, 2018).

In recent years, hybrid thermal management systems have emerged as an innovative approach that combines the advantages of both passive and active strategies. Notably, PV/T-PCM (Photovoltaic/Thermal systems with Phase Change Material) configurations have demonstrated the ability to moderate PV cell temperatures while simultaneously storing thermal energy (Şimşek and Ökten, 2025). These systems typically integrate a coolant channel beneath the PV cells and a PCM layer underneath the fluid domain, thus enhancing the heat extraction from the PV module. Studies have reported improvements in electrical efficiency of up to 3.8% and gains in thermal performance exceeding 7% under specific flow and irradiance conditions. Moreover, parametric analyses show that PV/T-PCM systems are particularly beneficial under moderate coolant flow rates, elevated ambient temperatures, and varying solar concentration ratios, offering a more robust performance envelope compared to conventional PV/T or passive-only systems.

Alongside geometrical and material innovations, the application of nanofluids has emerged as a powerful method to enhance thermal conductivity and convective heat transfer performance of cooling systems in solar applications. Nanofluids—engineered suspensions of nanoparticles in base fluids—exhibit superior thermal properties compared to conventional fluids, depending on the nanoparticle material, concentration, morphology, and fluid-nanoparticle interaction mechanisms (Choi et al., 2001; Eastman et al., 2001). Among the various nanomaterials investigated, silver-based nanofluids have attracted particular attention due to the exceptional characteristics of silver nanoparticles (AgNPs), including high thermal conductivity and chemical stability (Xu et al., 2020).

Recent developments in Hao et al. (2023) demonstrated that silver nanofluids exhibited high stability and uniform dispersion, with spherical nanoparticles averaging 8 nm in diameter. Thermal conductivity measurements revealed enhancements of up to 18.3% at 55 °C compared to the base fluid, with performance strongly correlated to nanoparticle concentration and synthesis duration. Moreover, the morphology of silver nanostructures plays a pivotal role in determining the heat transfer capabilities of the resulting nanofluids. Zhang et al. (2017) showed that silver nanowires (AgNWs) with high aspect ratios significantly outperformed spherical particles in enhancing thermal conductivity. In their experiments, nanofluids containing only 0.46 vol.% AgNWs achieved thermal conductivity gains exceeding 13%, attributed to the formation of interconnected thermal pathways and improved phonon transport efficiency. In light of these advances, the integration of Ag-based nanofluids into active cooling systems for high-concentration photovoltaic (HCPV) modules offers a promising strategy for improving heat dissipation. This, in turn, contributes to the stabilization of PV cell temperature and enhancement of electrical conversion efficiency, particularly under intense solar irradiance. Despite these promising developments, a comprehensive experimental comparison between passive and active cooling systems under real operating conditions—specifically tailored for HCPV applications—is still lacking in literature. Furthermore, the synergistic effects

of nanofluid properties, cooling system architecture, and thermal load intensity require deeper investigation to guide practical implementation.

This study addresses these research gaps by presenting a side-by-side evaluation of finned heat sinks (passive) and microchannel heat exchangers (active) applied to high-concentration photovoltaic modules. Active cooling system has been tested using standard base fluids (distilled water and a 50:50 mixture of water and ethylene glycol) and silver nanofluids with and without surfactants. Experimental data were collected under controlled laboratory conditions and validated in a real HCPV installation. Key performance metrics, including thermal resistance, heat transfer coefficient, pressure drop, and electrical efficiency were analyzed to provide a robust assessment of each cooling strategy's applicability and effectiveness.

By integrating insights from recent literature with experimental findings, this work contributes to the advancement of high-efficiency, thermally stable, and scalable cooling technologies for high-concentration solar energy systems. The results offer practical guidance for designers, researchers, and engineers working to improve the reliability and output of HCPV technologies.

Methodology

To evaluate the thermal and electrical performance of high-concentration photovoltaic (HCPV) modules under different cooling strategies, two distinct systems were designed, constructed, and experimentally analyzed: a passive aluminum finned heat sink and an active microchannel heat exchanger operating with silver-based nanofluids. The experimental methodology encompassed the synthesis and characterization of nanofluids, thermal performance testing under controlled heat flux, and in-situ validation using a prototype HCPV system installed outdoors.

Cooling Systems

Passive Cooling: Finned Heat Sink. The passive cooling system consisted of a rectangular aluminum heat sink with longitudinal fins optimized for natural convection. The device had the dimensions shown in Fig.1, and was manufactured from 6061 aluminum alloy, selected for its high thermal conductivity. Thermal contact between the HCPV cell and the heat sink was ensured by a thermal interface material (TIM) composed of silicone-based paste with thermal conductivity of $5.6 \text{ W/m}\cdot\text{K}$. Temperature distribution on the heat sink surface was monitored using T-type thermocouples, and thermal performance was assessed under both laboratory-controlled heating and real solar exposure

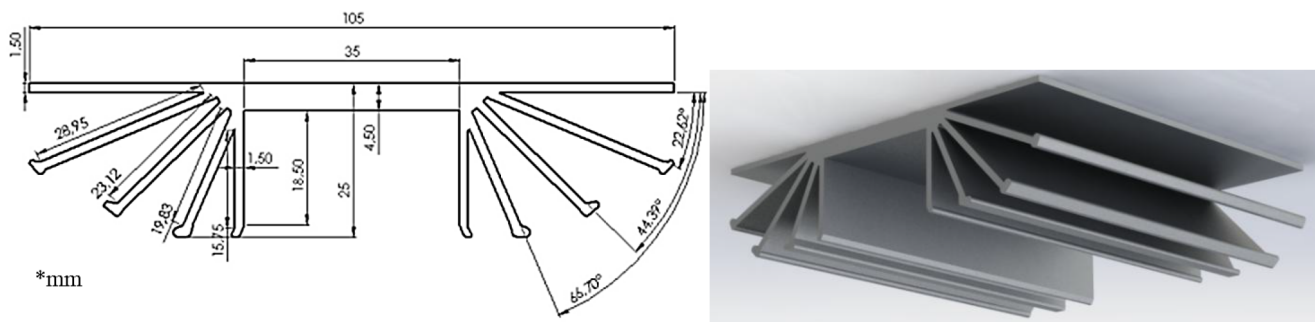


Figure 1—Geometrical characteristics of passive cooling finned heat sink.

Active Cooling System: Microchannel Heat Exchanger. In this study was used the microchannel heat exchanger designed by ((FERNANDES DE SOUSA, 2017) with an array of 21 parallel rectangular channels, each with width of $400 \mu\text{m}$ and a height of $900 \mu\text{m}$. The heat exchanger body was fabricated from aluminum to maximize heat transfer and minimize thermal resistance. The channels were enclosed using an aluminum cover and polyamine gaskets. Thermocouples and pressure transducers were integrated at the

inlet and outlet ports to record fluid temperature and pressure drop. A recirculating pump system with a flow controller enabled precise modulation of flow rates between 15 and 60 mL/min.

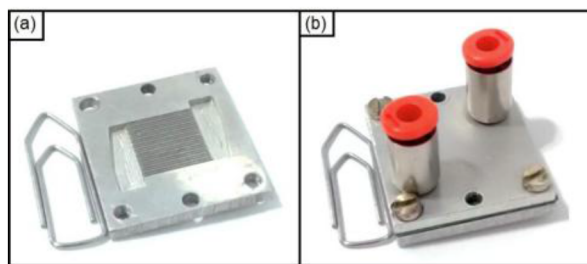


Figure 2—Microchannel heat exchanger used in this study: (a) base. (b) Sealed micro exchanger with hose connections. Source: (FERNANDES DE SOUSA, 2017)

Nanofluid Preparation and Characterization

Silver nanofluids were synthesized in two base fluids: distilled water (H_2O) and a 50:50 (wt%) mixture of H_2O and ethylene glycol (EG). The silver nanoparticles had nominal diameters of 20 nm and 80 nm, with and without polyvinylpyrrolidone (PVP) surfactant. Ultrasonication using a high-frequency probe was employed to ensure dispersion. Gravimetric methods confirmed that volume fractions of 0.0005%, 0.001%, and 0.005% were successfully obtained across 18 samples. Thermophysical properties, including thermal conductivity, dynamic viscosity, specific heat, and density—were experimentally measured using laboratory instrumentation (e.g., Brookfield viscometer, transient hot-wire technique) following established procedures.

Laboratory Experimental Setup

A test bench was constructed to simulate thermal loads representative of HCPV systems. Instead of real solar cells, a copper heating block incorporating a 40 W, 12 V cartridge heater was used to deliver uniform heat to the cooling devices. The heater was powered by a Minipa MPC-3006D DC source and monitored via Agilent 34411A multimeters. Temperature measurements were acquired using calibrated thermocouples (type K), and flow rates were controlled by a US Solar Pump DC gear pump integrated with a PID control loop. Flow measurement was ensured with a turbine flowmeter (FLR1007, Omega), and all data were acquired via a LabVIEW-controlled NI DAQ system.

Small Scale HCPV pilot System

The HCPV prototype used in this study comprised a high-efficiency multijunction photovoltaic cell (Azur Space, 41.4% nominal efficiency at 500 suns and 25°C) integrated into a custom module structure. Optical concentration was achieved using a 5×4 Fresnel lens array (Leutz Optics and Illumination UG) with a total collection area of 790 × 640 mm². Under ideal alignment, this configuration achieved an optical concentration ratio of approximately 744X. The module was mounted on a two-axis solar tracker composed of a steel pedestal with anticorrosive coating. As depicted in Fig. 3, two 24 V DC motors controlled by a feedback loop using a solar position sensor provided azimuth and elevation movement. The mechanical structure was designed to accommodate both passive and active cooling devices using standardized M3 fasteners. Temperature measurements inside the module were made with calibrated T-type thermocouples positioned at the inlet, middle, and outlet of each vertical cell column. Data acquisition was performed with a NI9213 module and NI9172 chassis. Field tests were conducted over 30 days, with environmental parameters recorded by a nearby meteorological station. These included solar irradiance, wind speed, and ambient temperature.

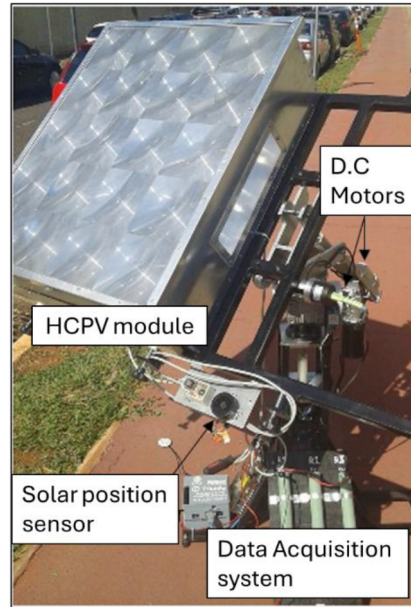


Figure 3—Outdoor HCPV system.

Evaluation Metrics

The performance of the passive and active cooling systems was assessed using thermodynamic and thermo-hydraulic indicators derived from experimental data. One of the primary quantities analyzed was the thermal resistance (R_{th}), calculated as the ratio of the temperature difference between the heat source and the coolant to the applied heat flux, as shown in Equation (1).

$$R_{th} = LMTD / q'' \quad (1)$$

Here, LMTD is the logarithmic mean temperature between the base temperature at the top surface of the micro exchanger (T_{bs}) and the fluid temperature at the inlet (T_{in}) and outlet (T_{out}) of the micro heat exchanger, and q'' is the applied surface heat flux.

The convective heat transfer coefficient (h) was obtained by rearranging Newton's law of cooling, as expressed in Equation (2), where Aw is the wetted area of the microchannels, where T_s is the surface temperature of the heated plate and T_f is the average temperature of the working fluid.

$$h = q'' / Aw (T_s - T_f) \quad (2)$$

The dimensionless Nusselt number (Nu), defined in Equation (3), was used to evaluate the convective performance in the microchannel domain.

$$Nu = h D_h / k \quad (3)$$

In this expression, D_h represents the hydraulic diameter of the channel and k is the thermal conductivity of the fluid. Hydraulic losses were characterized by the Darcy friction factor (f), which was computed using the pressure drop measurements and Equation (4), where L is the channel length, ρ is the fluid density, and V is the mean velocity of the fluid.

$$f = (2 D_h \Delta P) / (L \rho V^2) \quad (4)$$

The overall thermal-hydraulic benefit of nanofluids was assessed using the Performance Evaluation Criterion (PEC), as given in Equation (5).

$$PEC = (Nu / Nu_0) / ((f / f_0)^{1/3}) \quad (5)$$

In this expression, Nu and f refer to the nanofluid condition, while Nu_0 and f_0 represent the reference base fluid case. Values of PEC greater than 1 indicate that the use of nanofluids leads to a net performance improvement.

All experiments were repeated at least three times to ensure reproducibility. Measurement uncertainties were estimated using the root-sum-square method in accordance with GUM. The maximum uncertainty in calculated thermal resistance was below 5%, while temperature and flow measurement uncertainties were within $\pm 0.3^\circ\text{C}$ and $\pm 1.5\%$, respectively.

Results and discussions

Passive Cooling Results

The performance of the passive cooling system (an aluminum finned heat sink) was evaluated under both laboratory and outdoor conditions to assess its ability to dissipate heat from HCPV module. In the laboratory tests, the system was exposed to controlled heat fluxes of approximately 450, 670, 910, and 1150 W/m^2 , while maintaining ambient temperature at 25°C . The primary indicator of performance, thermal resistance, was calculated based on measured temperature differentials across the dissipator. Results in Fig. 4a and Fig. 4b, showed that the thermal resistance in steady-state conditions averaged 3.06°C/W , which aligns closely with the manufacturer's nominal value of 2.47°C/W . Deviations between theoretical and experimental results remained within 5.25%.

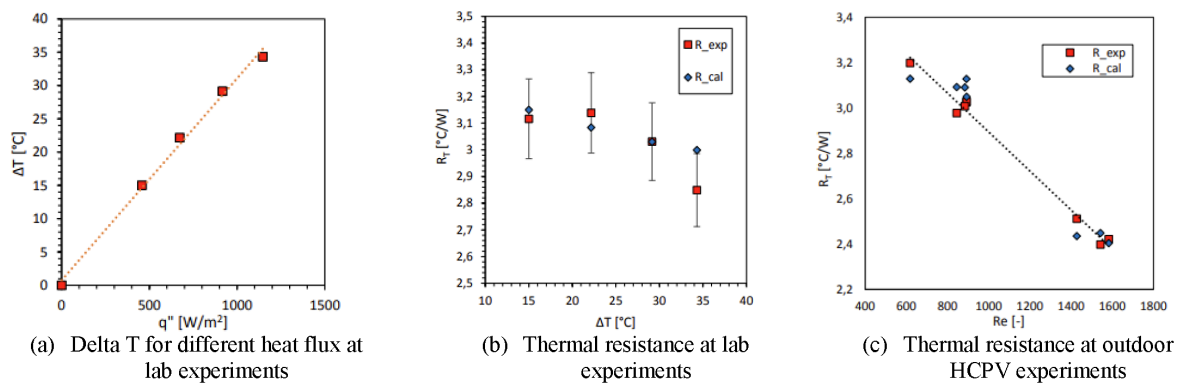


Figure 4—Passive cooling results.

In outdoor tests (Fig. 4c), the passive cooling system was integrated with an HCPV prototype, and thermal behavior was evaluated under real solar irradiance and wind conditions. To accurately model heat transfer under these variable conditions, a mixed convection analysis was adopted using the Richardson number. The inclusion of forced convection components, albeit at low Reynolds numbers (<1600), was justified based on validation with classical models from Maughan and Incropera (1990). Experimental thermal resistance values of 3.09°C/W (without mixed convection) and 2.32°C/W (with mixed convection) were recorded, demonstrating consistency with theoretical predictions and underscoring the influence of wind-assisted convection.

Furthermore, energy balance calculations between the measured and modeled heat dissipation showed average deviations of just 1.57%, validating the accuracy of the thermal model employed. The role of radiation heat transfer was also quantified, contributing less than 9% of total heat dissipation. It was confirmed that the finned structure dissipated approximately 86% of the heat via natural convection through its fins, with the remainder through the base plate and radiation.

Active Cooling Results

The microchannel heat exchanger was evaluated under a range of heat fluxes (7 to 22 kW/m²), inlet temperatures (25°C and 35°C), and volumetric flow rates (15 to 60 mL/min). The objective was to assess the capacity of the system to dissipate heat efficiently under laminar flow conditions, with distilled water and silver nanofluids as working fluids.

Initial validation of the experimental setup using distilled water showed excellent agreement with classical Nusselt number correlations for rectangular channels. At Reynolds numbers up to 2000, experimental Nusselt numbers averaged 3.67, consistent with theoretical values for thermally developed laminar flow with constant wall heat flux.

As shown in Fig.5a, the calculated heat transfer coefficients (h) for distilled water ranged from 1540 to 5680 W/m²·K, increasing with flow rate and decreasing slightly with inlet temperature. The pressure drop across the microchannel in Fig.5b, remained within acceptable limits (≤ 500 Pa), confirming the low hydraulic penalty of the configuration. Corresponding thermal resistance (R_{th}) values (Fig.5c) decreased from 0.76°C/W to 0.25°C/W as flow increased from 15 to 60 mL/min at 25°C inlet temperature.

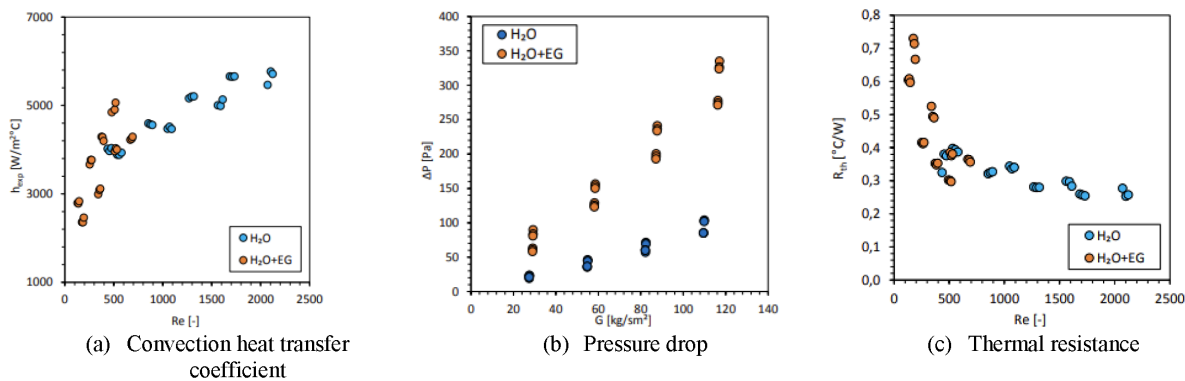


Figure 5—Active cooling results for microchannel heat sink

Comparative Analysis of Passive vs. Active Cooling in HCPV Systems

To evaluate the real-world effectiveness of the heat exchanger, an energy balance analysis was conducted considering the results obtained at outdoor conditions in the finned heat sink analysis. The performance of the passive cooling configuration was evaluated in clear sky conditions with global solar irradiance (G) between 800.11 W/m² and 911.7 W/m². The most critical parameter in this context is the solar cell temperature (T_{cell}), which reached values as high as 66.95°C and remained consistently above 60°C in almost all scenarios. Corresponding system electrical efficiencies (η_{el}) varied from 26.77% to 31.02%, demonstrating sensitivity to both irradiance level and module temperature.

Table 1 presents the estimated results for the HCPV system employing active cooling. Under identical operating conditions, the passive cooling configuration yielded a cell power output of 5.43 W, a solar cell temperature of 67.6°C, and an electrical efficiency of 28.82%. In contrast, the use of active cooling with microchannel heat exchangers limited the maximum cell temperature to 49.99°C, with a further reduction to 45.21°C at higher coolant flow rates. This corresponds to a temperature decrease of up to 21.7%, which is both thermally and electrically significant. However, this thermal improvement did not result in a clearly observable effect on the electrical efficiency trend.

Table 1—Efficiency analysis on HCPV module for the microchannel heat exchanger

$\dot{V}$ [mL/min]	T_{cel} [°C]	$\dot{Q}_f$ [W]	P_{cell} [W]	$\dot{W}_p$ [mW]	η_{th} [%]	η_{ele} [%]	η_G [%]
15	49.99	8.43	5.52	0.14	45.0	29.45	74.45
30	46.84	8.54	5.54	0.43	45.55	29.57	75.12
45	45.95	8.94	5.55	1.09	47.68	29.61	77.28
60	45.21	9.21	5.56	2.05	49.16	29.64	78.79

Thermal efficiency (η_{th}) also improved with higher flow rates, increasing from 45.0% to 49.16%, confirming that more heat was successfully removed and transferred to the fluid. Even when accounting for the pumping power expenditure ($\dot{W}_p$), which rose from 0.14 mW to 2.05 mW, the gain in thermal performance more than compensates for the slight increase in auxiliary consumption. This is evident in the global efficiency (η_G), which increased from 74.45% to 78.79%, surpassing typical values associated with passive configurations, which often remain below 75%.

Nanofluid Properties and Thermohydraulic Performance

The thermophysical characterization of silver-based nanofluids was conducted to evaluate their suitability for integration into microchannel heat exchangers for HCPV thermal management. The nanofluids were formulated using distilled water and a 50:50 weight mixture of distilled water and ethylene glycol (H₂O–EG) as base fluids, with silver nanoparticles of different diameters (20 nm and 80 nm), with and without polyvinylpyrrolidone (PVP) surfactant. Three volumetric concentrations were studied: $\Phi_1 = 0.0005\%$, $\Phi_2 = 0.001\%$, and $\Phi_3 = 0.005\%$.

Measurements of dynamic viscosity, density, thermal conductivity, and specific heat were carried out over a temperature range of 25°C to 55°C. Viscosity and density decreased with increasing temperature, with the dynamic viscosity exhibiting exponential decay, while the density showed a linear trend. The average relative increases in dynamic viscosity for the H₂O–EG-based nanofluids were modest, not exceeding 2.33%, while relative density changes remained below 0.5% for all samples, indicating manageable hydraulic penalties.

Thermal conductivity, on the other hand, increased with both nanoparticle concentration and temperature, especially for the Ag_80 and Ag_20 formulations. The Ag_80 nanofluid at the highest concentration (Φ_3) showed a 4.08% increase in thermal conductivity compared to the base fluid, while Ag_20 exhibited a 3.28% enhancement. Interestingly, the Ag_PVP nanofluid displayed its highest thermal conductivity gain at the lowest concentration tested (Φ_2), indicating a non-linear relationship between particle loading and heat transfer potential, likely due to surfactant-induced dispersion stability and interfacial thermal resistance effects.

The specific heat of the nanofluids decreased slightly with increasing nanoparticle content, as expected. All samples showed reduced heat capacity relative to the base fluids, with maximum reductions near 4.7%, in agreement with theoretical models assuming homogeneous two-phase mixtures. However, these reductions did not outweigh the gains in convective heat transfer observed in microchannel experiments.

Figures 6 and 7 shown the thermohydraulic performance that was assessed using dimensionless analysis comparing the relative heat transfer coefficient ($h_{\text{nf}}/h_{\text{bf}}$) and pumping power ($\dot{W}_{\text{P,nf}}/\dot{W}_{\text{P,bf}}$). Results were plotted in normalized performance maps based on the methodology by Cárdenas (2019), with four performance quadrants: desirable (Q1), expected (Q2), undesirable (Q3), and unfavorable (Q4). The Ag_80 samples tended to cluster in Q2, reflecting increased heat transfer but at higher pumping cost, while the Ag_PVP samples performed better overall, particularly at $\Phi_2 = 0.001\%$, where results fell in Q1, indicating both thermal enhancement and reduced hydraulic penalty. In contrast, nanofluids with 20 nm Ag nanoparticles without PVP showed poor dispersion stability and inconsistent performance, with many

operating points falling in Q3, indicating simultaneous losses in heat transfer and increased pressure drop. These fluids were ultimately excluded from further analysis due to rapid sedimentation during extended testing.

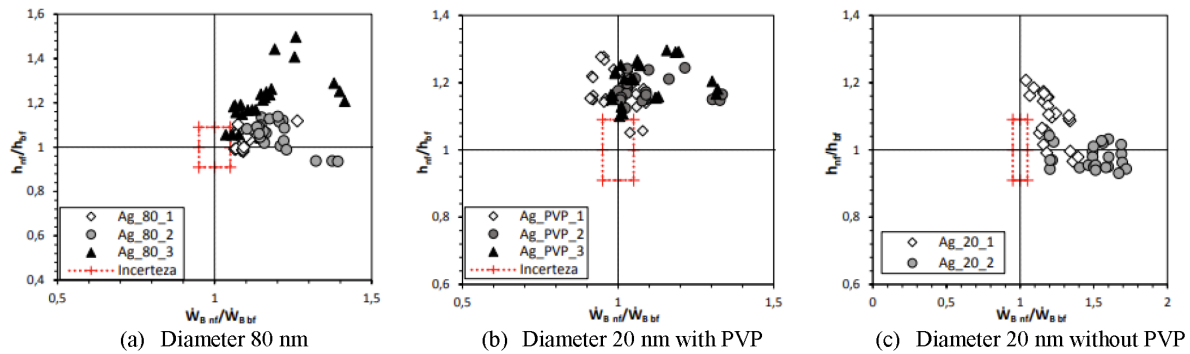


Figure 6—Thermohydraulic performance of the different H₂O+EG nanofluids flowing in microchannel heat sink

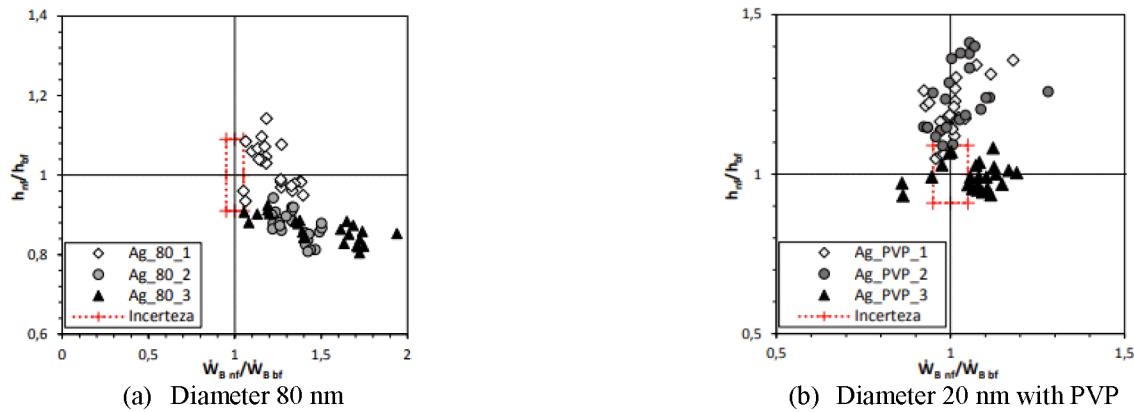


Figure 7—Thermohydraulic performance of the different H₂O nanofluids flowing in microchannel heat sink

In terms of thermal resistance silver-based nanofluids demonstrated marked improvements over conventional base fluids. Specifically, when compared to a mixture of distilled water and ethylene glycol, the nanofluids achieved average reductions in thermal resistance of up to 17.09% for Ag_20 with PVP and 18.3% for Ag_80, particularly at higher nanoparticle concentrations (Φ_3). This reduction in thermal resistance suggests the potential for enhanced electrical and thermal efficiency when applied to high-concentration photovoltaic (HCPV) systems. These findings highlight the strong potential of such cooling technologies for applications involving high heat fluxes and advanced thermal management requirements.

Conclusions

This study presented a comparative experimental analysis of passive and active thermal management strategies applied to high-concentration photovoltaic (HCPV) modules. Two systems were evaluated: a finned aluminum heat sink and a microchannel heat exchanger operating with base fluids and silver-based nanofluids.

The passive cooling configuration, while mechanically simple and energy-independent, exhibited thermal limitations under outdoor conditions. The thermal resistance exceeded 3°C/W and the maximum cell temperature surpassed 66°C, resulting in a constrained electrical efficiency range (26.8–31%). These results highlight the restricted capacity of natural convection and radiative mechanisms to dissipate concentrated solar thermal loads effectively.

In contrast, the active microchannel system demonstrated superior thermal regulation capabilities. With base fluids, thermal resistance decreased to 0.25°C/W at higher flow rates, and the cell temperature was reduced by over 17°C compared to the passive setup. This translated to a measurable improvement in global efficiency, rising from 74.45% to 78.79%, with electrical efficiencies reaching up to 29.64%.

The integration of silver-based nanofluids further enhanced the performance of the active system. Nanofluids prepared with PVP-stabilized Ag nanoparticles achieved thermal resistance reductions of up to 18.3% over the base fluid, while maintaining low viscosity penalties. The best-performing nanofluids delivered improved heat transfer coefficients and shifted operation into the "desirable" thermohydraulic quadrant (Q1), combining efficiency gains with minimal hydraulic losses.

Overall, the results confirm that nanofluid-enhanced microchannel cooling offers a promising path for improving thermal and electrical performance in HCPV systems. The combined benefits of reduced cell temperature, increased heat extraction, and stabilized output under fluctuating solar conditions support its integration into high-efficiency, scalable solar energy architectures.

Acknowledgement

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